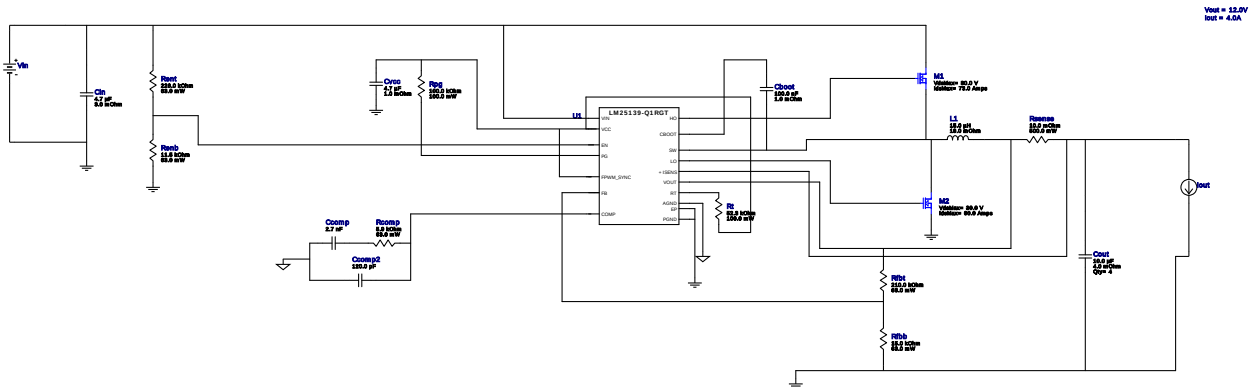




Design : 13 LM25139QRGTRQ1
LM25139QRGTRQ1 23V-24V to 12.00V @ 4A

VinMin = 23.0V
VinMax = 24.0V
Vout = 12.0V
Iout = 4.0A

Device = LM25139QRGTRQ1
Topology = Buck
Created = 2026-03-28 08:48:47.775
BOM Cost = \$2.57
BOM Count = 21
Total Pd = 1.12W



Design Alerts

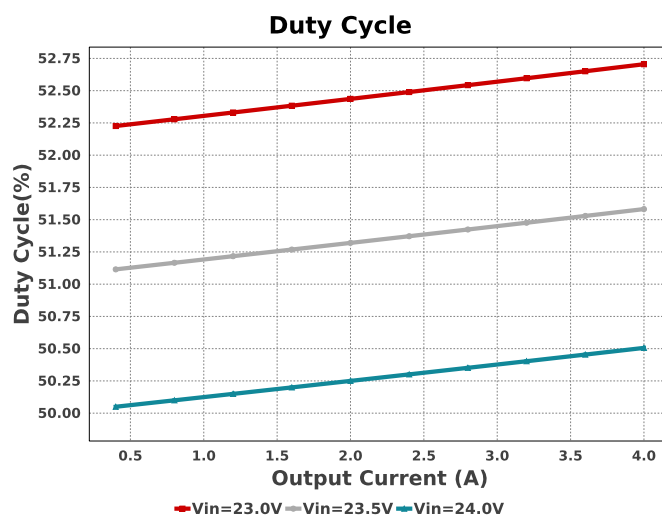
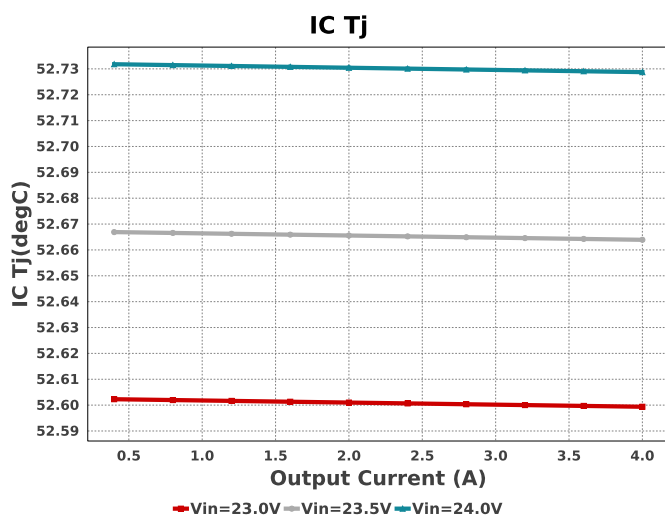
Component Selection Information

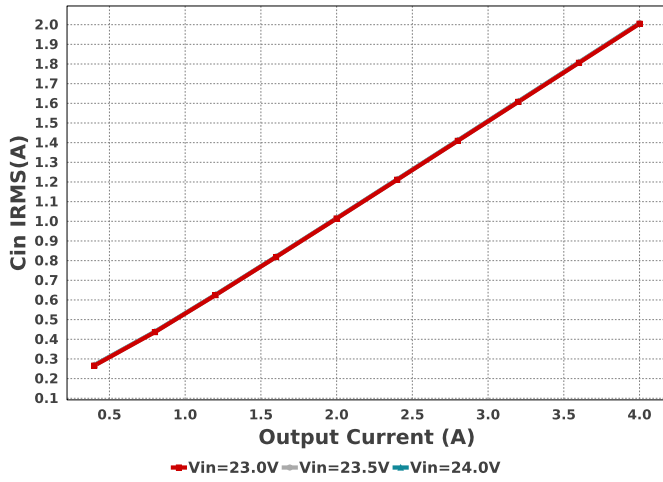
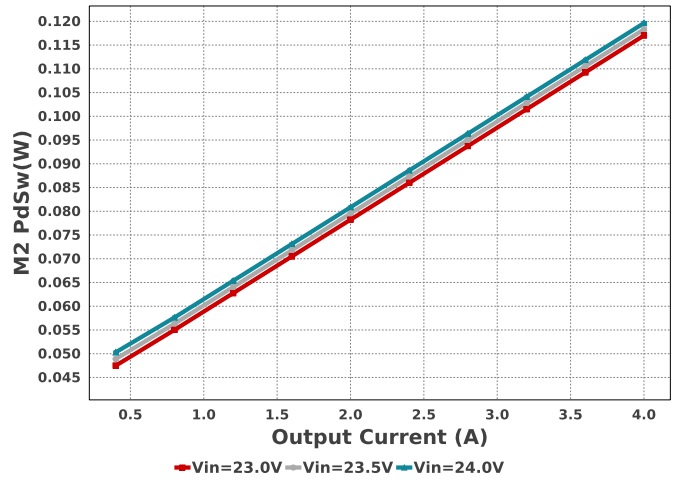
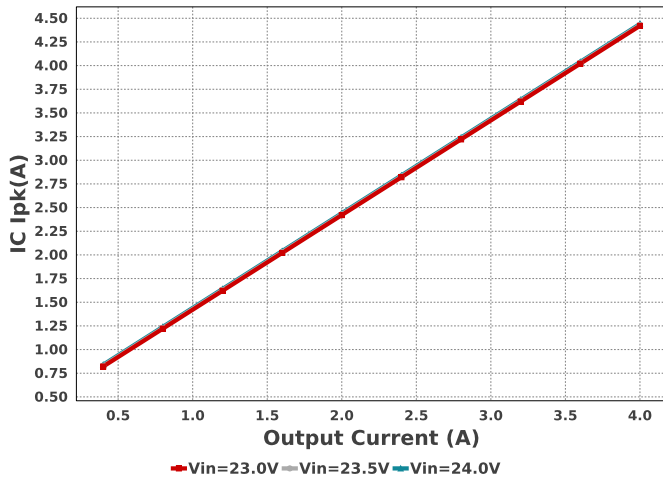
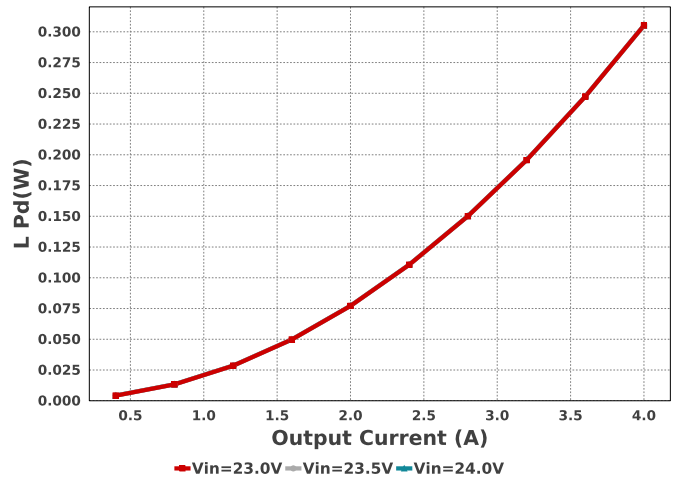
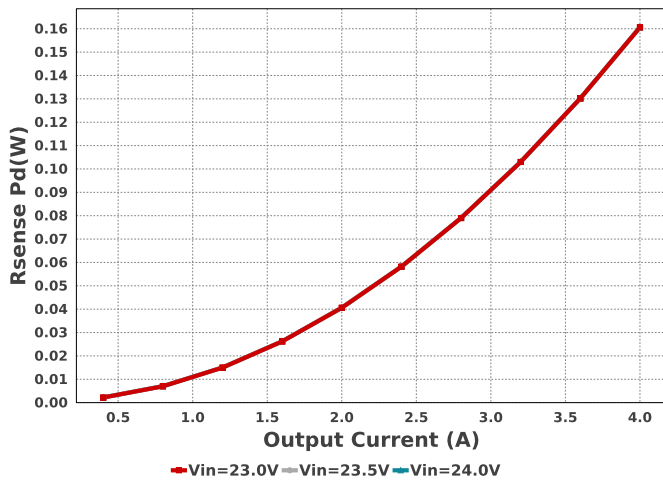
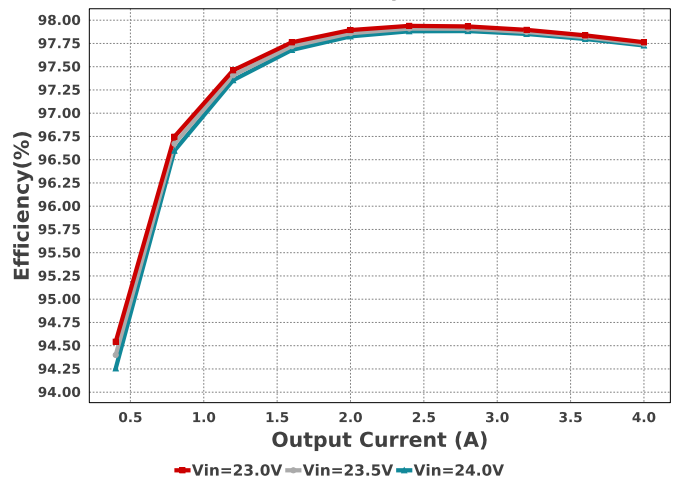
The LM25139-Q1 is qualified for Automotive applications. All passives and other components selected in this design may not be qualified for Automotive applications. The user is required to verify that all components in the design meet the qualification and safety requirements for their specific application. Please enter Enable UVLO Start Threshold greater than Enable 1.1* UVLO Stop Threshold. UVLO Start Threshold is assumed as 23.0V to avoid invalid results.

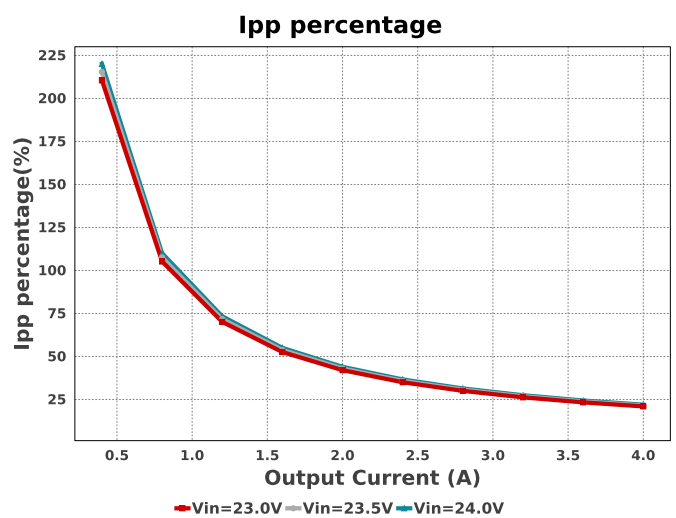
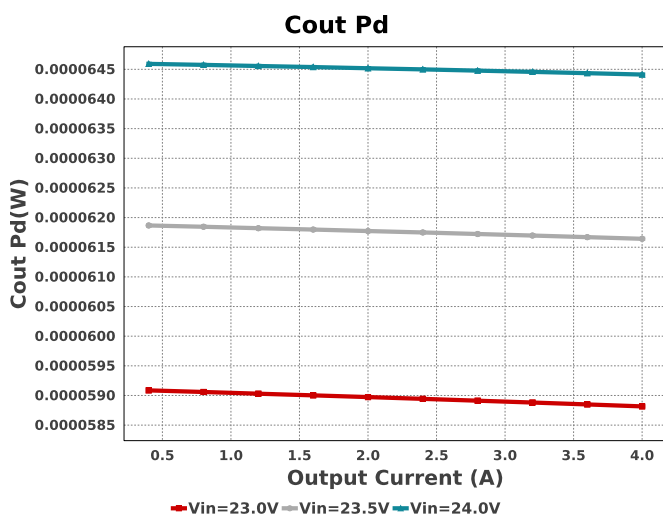
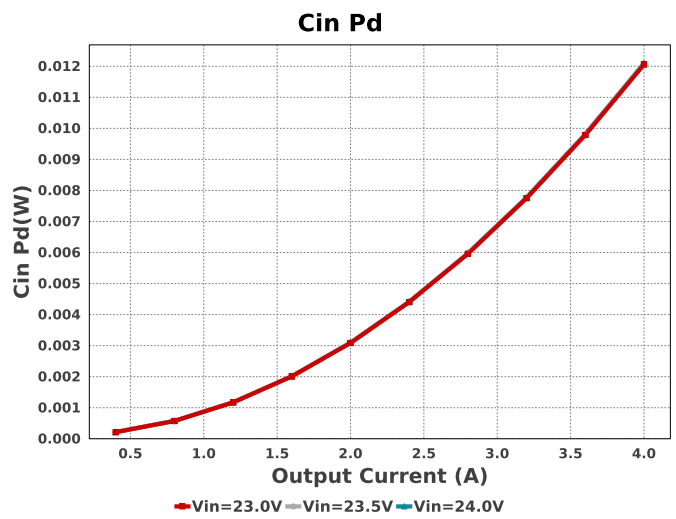
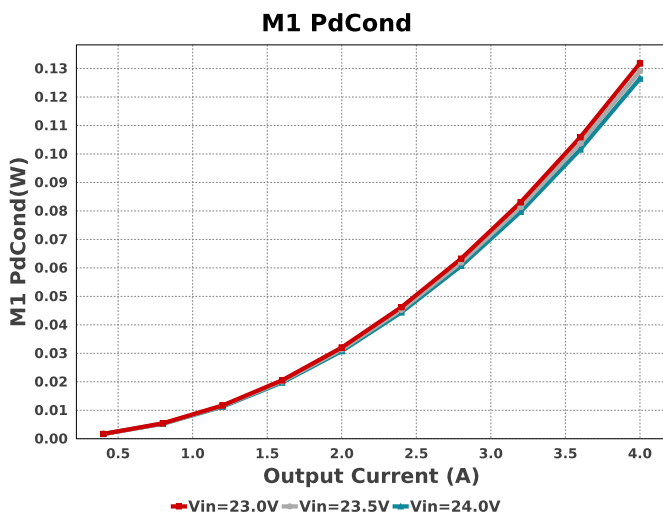
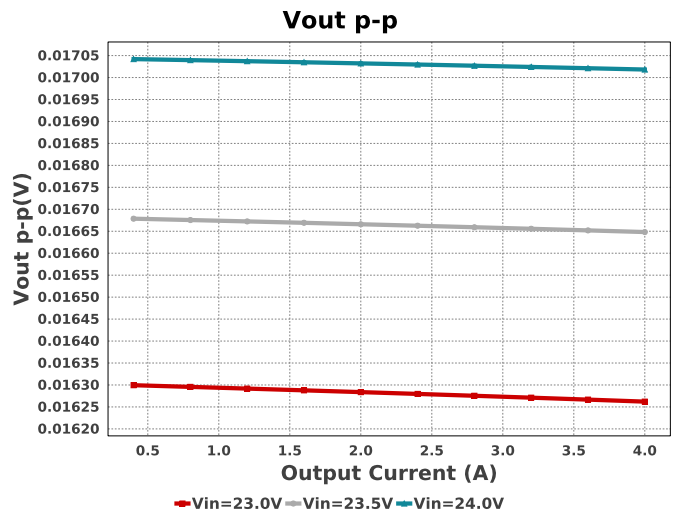
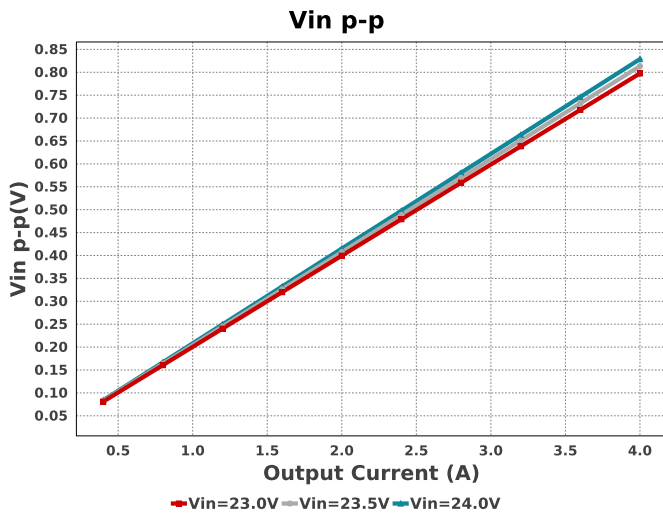
Electrical BOM

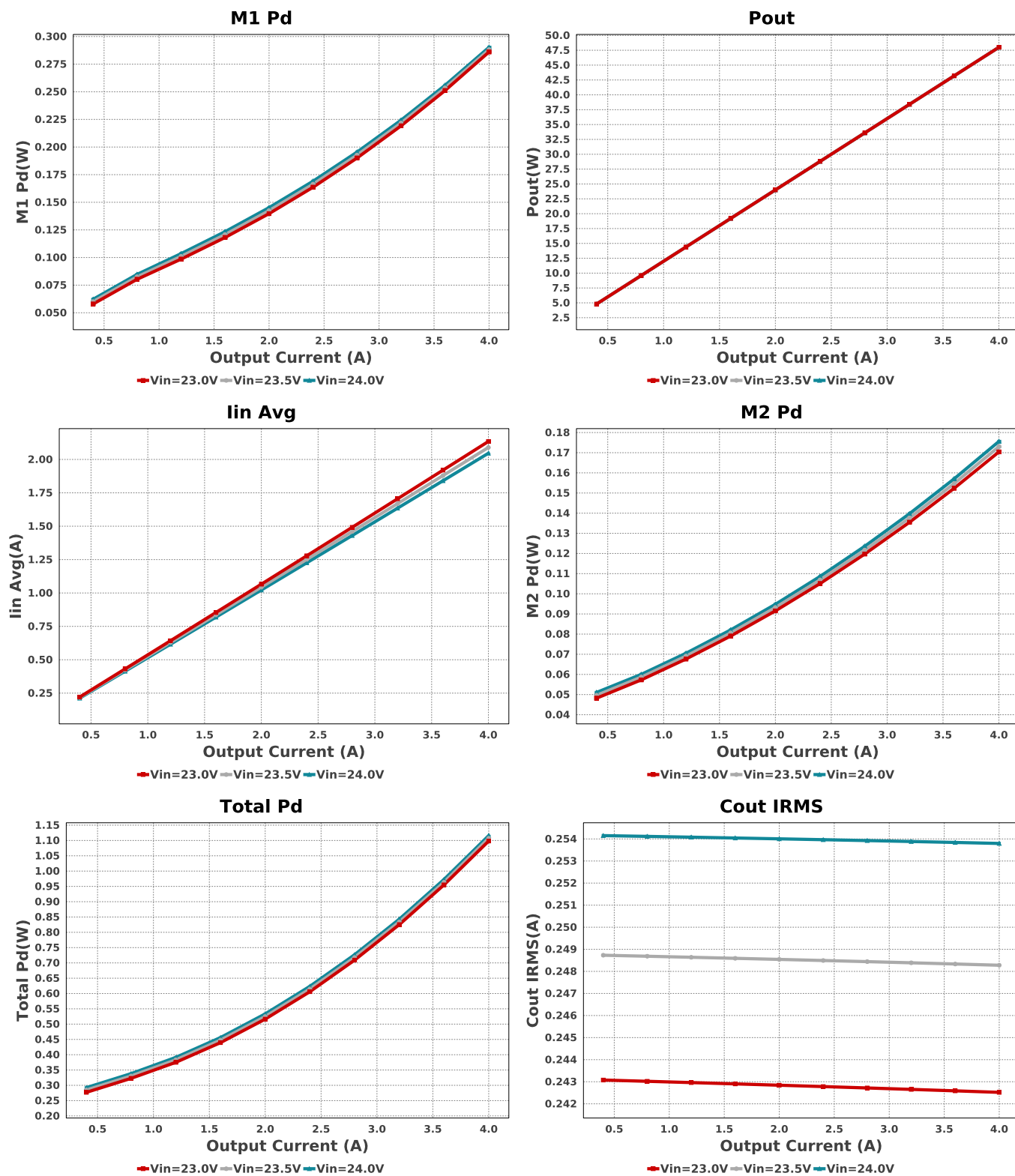
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71A104KA01D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	 0402 3 mm ²
Ccomp	MuRata	GRM1885C1H272JA01J Series= C0G/NP0	Cap= 2.7 nF VDC= 50.0 V IRMS= 0.0 A	1	\$0.02	 0603 5 mm ²
Ccomp2	Taiyo Yuden	UMK105CG121JV-F Series= C0G/NP0	Cap= 120.0 pF VDC= 50.0 V IRMS= 0.0 A	1	\$0.01	 0402 3 mm ²
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	1	\$0.10	 1206 11 mm ²
Cout	MuRata	GRM31CR71E106KA12L Series= X7R	Cap= 10.0 uF ESR= 4.0 mOhm VDC= 25.0 V IRMS= 6.0 A	4	\$0.06	 1206_180 11 mm ²
Cvcc	Taiyo Yuden	LMK212B7475KG-T Series= X7R	Cap= 4.7 uF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.04	 0805 7 mm ²
L1	Coilcraft	MSS1210-153MEB	L= 15.0 uH 19.0 mOhm	1	\$0.81	 MSS1210 204 mm ²

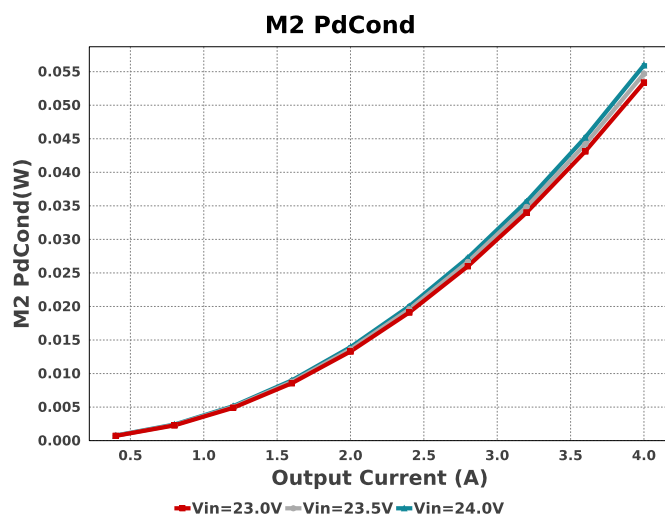
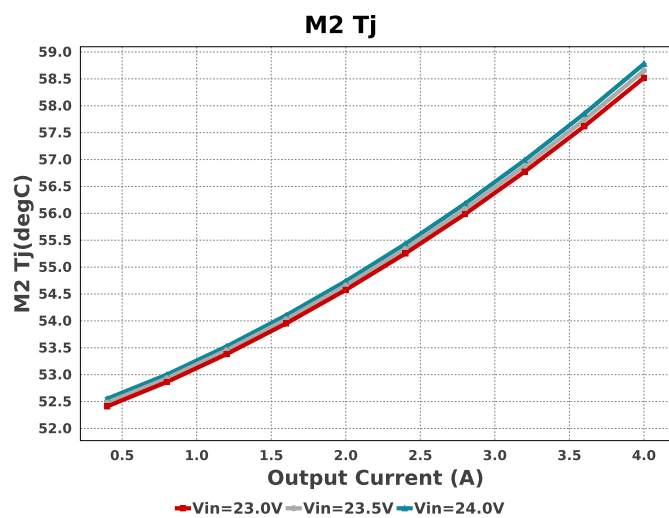
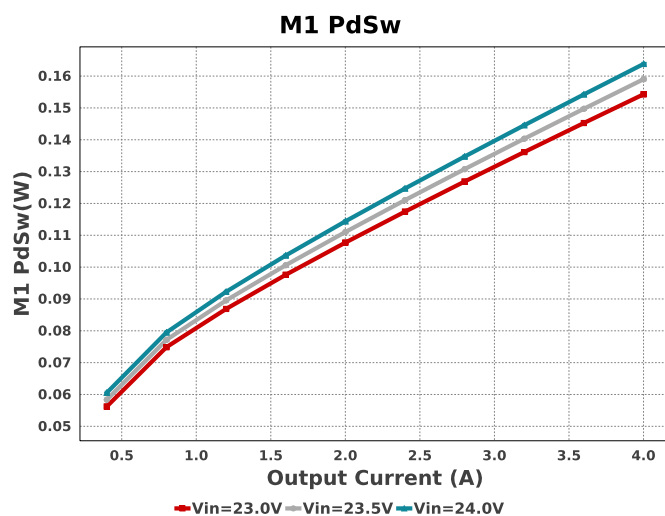
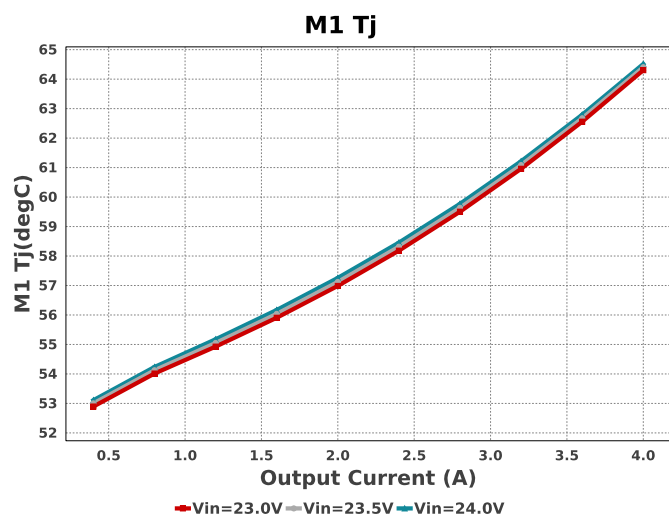
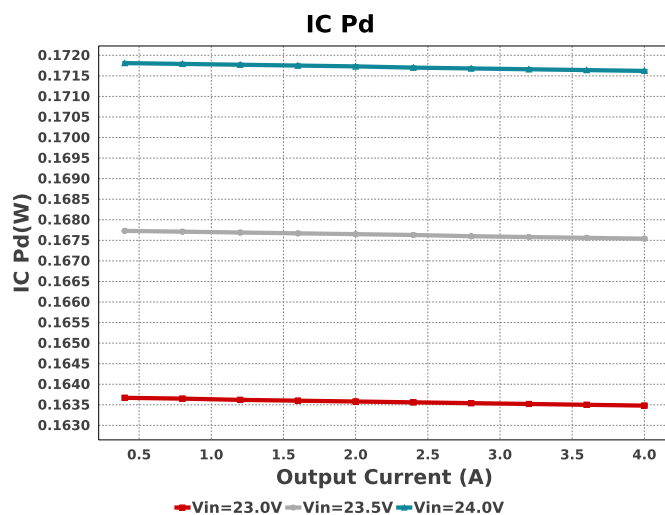
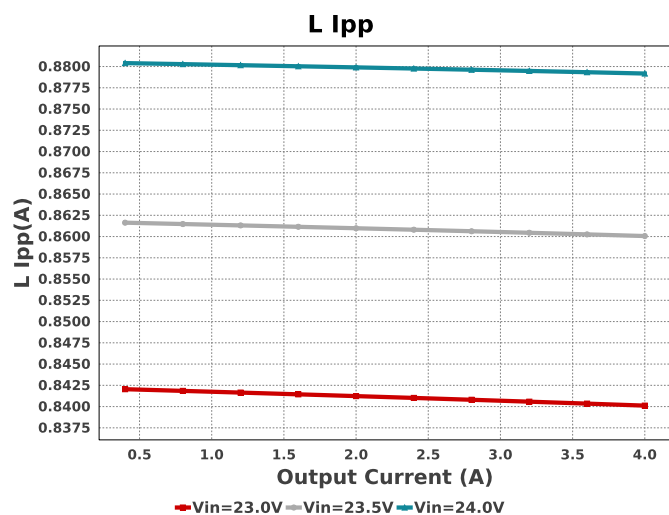
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
M1	Texas Instruments	CSD17307Q5A	VdsMax= 30.0 V IdsMax= 73.0 Amps	1	\$0.21	 TRANS_NexFET_Q5A 55 mm ²
M2	Texas Instruments	CSD17577Q5A	VdsMax= 30.0 V IdsMax= 60.0 Amps	1	\$0.17	 TRANS_NexFET_Q5A 55 mm ²
Rcomp	Vishay-Dale	CRCW04025K90FKED Series= CRCW..e3	Res= 5.9 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Renb	Vishay-Dale	CRCW040211K5FKED Series= CRCW..e3	Res= 11.5 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rent	Vishay-Dale	CRCW0402226KFKED Series= CRCW..e3	Res= 226.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfbb	Vishay-Dale	CRCW040215K0FKED Series= CRCW..e3	Res= 15.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rfbt	Vishay-Dale	CRCW0402210KFKED Series= CRCW..e3	Res= 210.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	 0402 3 mm ²
Rpg	Vishay-Dale	CRCW0603100KFKEA Series= CRCW..e3	Res= 100.0 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm ²
Rsense	Stackpole Electronics Inc	CSR1206FK10L0 Series= ?	Res= 10.0 mOhm Power= 500.0 mW Tolerance= 1.0%	1	\$0.11	 1206 11 mm ²
Rt	Vishay-Dale	CRCW060352K3FKEA Series= CRCW..e3	Res= 52.3 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	 0603 5 mm ²
U1	Texas Instruments	LM25139QRGTRQ1	Switcher	1	\$0.78	RGT0016K-MFG 16 mm ²

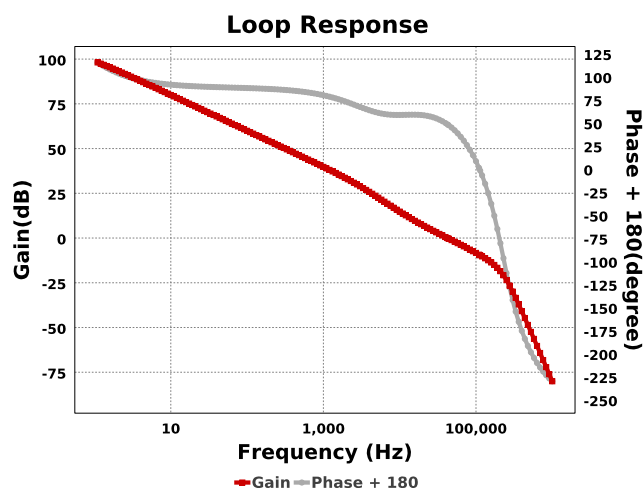


Cin IRMS**M2 PdSw****IC Ipk****L Pd****Rsense Pd****Efficiency**









Operating Values

#	Name	Value	Category	Description
1.	Cin IRMS	2.008 A	Capacitor	Input capacitor RMS ripple current
2.	Cin Pd	12.096 mW	Capacitor	Input capacitor power dissipation
3.	Cout IRMS	253.796 mA	Capacitor	Output capacitor RMS ripple current
4.	Cout Pd	64.412 μ W	Capacitor	Output capacitor power dissipation
5.	IC Ipk	4.44 A	IC	Peak switch current in IC
6.	IC Pd	171.62 mW	IC	IC power dissipation
7.	IC Tj	52.729 degC	IC	IC junction temperature
8.	IC Tolerance	10.0 mV	IC	IC Feedback Tolerance
9.	ICThetaJA Effective	15.9 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
10.	Iin Avg	2.046 A	IC	Average input current
11.	Ipp percentage	21.979 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
12.	L Ipp	879.18 mA	Inductor	Peak-to-peak inductor ripple current
13.	L Pd	305.22 mW	Inductor	Inductor power dissipation
14.	M1 Pd	290.28 mW	Mosfet	M1 MOSFET total power dissipation
15.	M1 PdCond	126.48 mW	Mosfet	M1 MOSFET conduction losses
16.	M1 PdSw	163.8 mW	Mosfet	M1 MOSFET switching losses
17.	M1 Tj	64.514 degC	Mosfet	M1 MOSFET junction temperature
18.	M2 Pd	175.58 mW	Mosfet	M2 MOSFET total power dissipation
19.	M2 PdCond	55.954 mW	Mosfet	M2 MOSFET conduction losses
20.	M2 PdSw	119.63 mW	Mosfet	M2 MOSFET switching losses
21.	M2 Tj	58.779 degC	Mosfet	M2 MOSFET junction temperature
22.	Cin Pd	12.096 mW	Power	Input capacitor power dissipation
23.	Cout Pd	64.412 μ W	Power	Output capacitor power dissipation
24.	IC Pd	171.62 mW	Power	IC power dissipation
25.	L Pd	305.22 mW	Power	Inductor power dissipation
26.	M1 Pd	290.28 mW	Power	M1 MOSFET total power dissipation
27.	M1 PdCond	126.48 mW	Power	M1 MOSFET conduction losses
28.	M1 PdSw	163.8 mW	Power	M1 MOSFET switching losses
29.	M2 Pd	175.58 mW	Power	M2 MOSFET total power dissipation
30.	M2 PdCond	55.954 mW	Power	M2 MOSFET conduction losses
31.	M2 PdSw	119.63 mW	Power	M2 MOSFET switching losses
32.	Rsense Pd	160.65 mW	Power	LED Current Rsns Power Dissipation
33.	Total Pd	1.116 W	Power	Total Power Dissipation
34.	Rsense Pd	160.65 mW	Resistor	LED Current Rsns Power Dissipation
35.	BOM Count	21	System	Total Design BOM count
36.	Cross Freq	42.315 kHz	System	Bode plot crossover frequency
37.	Duty Cycle	50.506 %	System	Duty cycle
38.	Efficiency	97.728 %	System	Steady state efficiency
39.	FootPrint	438.0 mm ²	System	Total Foot Print Area of BOM components
40.	Frequency	454.277 kHz	System	Switching frequency
41.	Gain Marg	-9.659 dB	System	Bode Plot Gain Margin
42.	Iout	4.0 A	System	Iout operating point

#	Name	Value	Category	Description
43.	Iout transient step used for Cout calculations	2.0 A	System Information	Custom Transient current step requirement that was used for Cout selection (A).
44.	Low Freq Gain	98.131 dB	System Information	Gain at 1Hz
45.	Mode	FCCM	System Information	Conduction Mode
46.	Overshoot Value	175.751 mV	System Information	Theoretical Vout Overshoot Value
47.	Phase Marg	50.087 deg	System Information	Bode Plot Phase Margin
48.	Pout	48.0 W	System Information	Total output power
49.	Total BOM	\$2.57	System Information	Total BOM Cost
50.	Undershoot Value	330.504 mV	System Information	Theoretical Vout Undershoot Value
51.	Vin	24.0 V	System Information	Vin operating point
52.	Vin p-p	828.784 mV	System Information	Peak-to-peak input voltage
53.	Vout	12.0 V	System Information	Operational Output Voltage
54.	Vout Actual	12.0 V	System Information	Vout Actual calculated based on selected voltage divider resistors
55.	Vout Ripple requirement used for Cout calculations	1.0 %	System Information	Custom maximum output ripple requirement that was used for Cout selection(% of Vout).
56.	Vout Tolerance	3.159 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
57.	Vout p-p	17.018 mV	System Information	Peak-to-peak output ripple voltage
58.	Vout transient requirement used for Cout calculations	3.0 %	System Information	Custom Transient voltage change requirement that was used for Cout selection (% of Vout).

Design Inputs

Name	Value	Description
Iout	4.0	Maximum Output Current
VinMax	24.0	Maximum input voltage
VinMin	23.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LM25139-Q1	Base Product Number
source	DC	Input Source Type
Ta	50.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

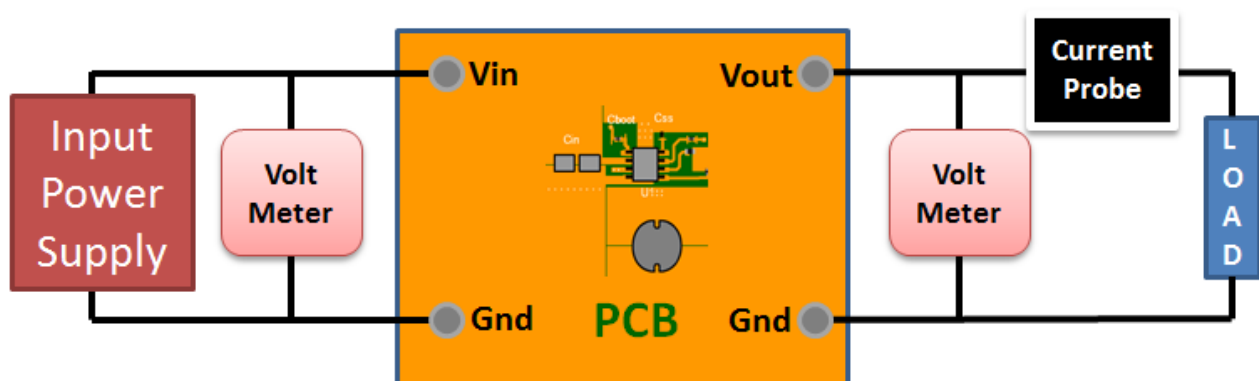
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 23.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



Design Assistance

1. Master key : 14CACC71A79F24D6AF9667573EFEAE90[v1]
2. **LM25139-Q1** Product Folder : <http://www.ti.com/product/LM25139%2DQ1> : contains the data sheet and other resources.

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